

Project Proposal

Project Name: STARS Autonomous Charging System

Project Team Members: John Ferguson, Evan Giordano, Josh Radjavitch

Problem Statement:

The problem to solve for our team is to effectively create some sort of charging area station(s) for the swarm robots to return to for autonomous battery charging during and after operational requirements of the individual bot to meet an ideal operating time. The goal of the charging system is to charge the whole swarm autonomously.

Why is it a problem?

What type of constraints the problem have? [eg: physical, cost, schedule, etc.]

The robots have a limited battery life, which reduces productivity and prevents individual units from completing their full operational tasks. This directly impacts the effectiveness of the entire swarm. If even one bot's battery dies, the mission fails, as uninterrupted operation with little to no human interaction as stated by the lead on the project ““The STARS project explicitly requires autonomous charging infrastructure to reduce human intervention and maximize resilience” (Kohlmeyer, 2025)” is a core customer requirement. As more bots are introduced, one of the most critical software challenges becomes the queuing algorithm that directs swarm members to the correct charging station at the right time. Physically, the bots currently lack effective charging infrastructure—such as Qi-based chargers—that can meet the individual charging needs of every unit. Additionally, the swarm is constrained by a limited arena size (8 feet by 10 feet), which will be scaled as more bots are deployed. From a cost–benefit perspective,

frequent robot failures due to power loss reduce overall output and force the purchase of larger batteries or reliance on single charging ports that require human oversight. Both options conflict with project goals. The customer explicitly demands that charging and scheduling processes remain fully autonomous, ensuring that the swarm operates continuously and efficiently with minimal to zero human intervention.

Stakeholders:

The primary stakeholders for this project are Dr. Watson, Jonah Kohlmeyer, and the BID4R Lab. These three entities combined oversee a project where a swarm of bots are set out to accomplish certain operational tasks. For this project to be successful, it is required to feature an autonomous charging system where the individual bots can be charged. Secondary stakeholders include people who would like to develop a charging system for other projects who could look at our solution to see if it is viable for whatever they are working on, as well as ERAU Research in general, as this project will draw more positive attention to research projects at ERAU. In addition, this project will be beneficial to the public at large, as it will be used for educational demonstrations in and outside of classrooms.

Proposed Solution (Scope):

The scope of our project covers the design and use of the charging stations as well as the receiver for the bots. It also includes the logic for the bots to decide who should be charging. The actual bots are already designed so they are out of scope.

Standards:

ERAU Safety Standards

OSHA

Qi Wireless Charging Standard

IEEE 802.15.4, 12207-2017, 1012, 1012:2016, 1201-2016, 2017

ISO 13482

ISO/IEC 29119

V&V

NFPA 855

Semester 1:

- *Utilize five swarm bots to be able to use one charging station while the bots that are not charging are operational for eight hours.*
- *One swarm bot can make a connection between the transmitter and receiver to charge.*
- *One swarm bot can autonomously reach a charging station and make connection to start charging.*
- *One swarm bot will leave the charging station once charged to the correct amount.*
- *Multiple swarm bots can queue to use one charging station.*

- Five swarm bots can queue to allow one bot to charge by battery percentage through one charging station to last for eight hours.

Semester 2:

- Increase the number of swarm bots to a maximum of 50 to allow for maximum operational time.

- Allow more bots to queue themselves based on battery percentage.

- Increase the amount of swarm bots that can be controlled at the same time.

- Increase the amount of charging stations.

- Allow bots to choose from different charging stations.

- Ability to keep the entire swarm charged and operational for 24 hours.

Proposed Project Budget:

The customer offered \$500 initially to support this project, so instead of \$1200 we will go with the former. Components include a charge transmitter and receiver. Ideally transmitters will handle multiple bots, but could be potential for each bot. There will be one receiver for each swarm bot. This will give us about \$8 to \$27 giving us only the ability to maintain about 15 bots. With this budget we can at least get the Semester One milestone of five bots. The scope of 50 bots depends on if the equipment we need per bot changes.

References:

APA 7th:

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- Kohlmeyer, J. (2025). *STARS autonomous charging system: Capstone proposal*. BID4R Lab, Embry-Riddle Aeronautical University
- International Organization for Standardization. (2014, February). *ISO 13482:2014 Robots and robotic devices — Safety requirements for personal care robots* (ISO Standard No. 13482:2014).

- National Fire Protection Association. (2020). NFPA 855: *Standard for the installation of stationary energy storage systems*. NFPA.

IEEE:

- IEEE, IEEE Standard for Software Life Cycle Processes, IEEE Std 12207-2017, 2017.
- IEEE, IEEE Standard for System, Software, and Hardware Verification and Validation, IEEE Std 1012-2016, 2016.
- IEEE, IEEE Standard for Verification and Validation of Hardware Elements of Digital Systems in Safety Systems, IEEE Std 1201-2016, 2016.
- R. R. Kale, “Development of a robotic swarm test arena for multi-agent resilient systems,” Graduate Research Project, M.S., Embry-Riddle Aeronautical University, Daytona Beach, FL, Spring 2025.